Question Paper

Back

# MANIPAL ACADEMY OF HIGHER EDUCATION

**MidTerm Examination,**
**School of Computer Engineering**

**DATA COMMUNICATION AND COMPUTER NETWORKS [CSS 2102]**

**Marks: 30** **Duration: 90 mins.**

**MCQ**

**Answer all the questions.** Section Duration: 20 mins

1) A 1000-byte frame is sent using Stop-and-Wait ARQ. The datarate is **1 Mbps**, and one-way propagation delay is **25 ms**. What is the channel utilization ? (0.5)

13.8 % 1.38 % 1.9% 19 %

2) In a communication system using Selective Repeat ARQ, the sequence number field is 8 bits long. To avoid ambiguity between old and new frames with the same sequence number, what is the maximum allowable sender window size? (0.5)

128 255 256 127

3) If an organization has 10 nodes, and wants to link the nodes using mesh topology, the number of links required for duplex communication is: (0.5)

10 45 90 20

4) Hop to Hop delivery is the responsibility of (0.5)

Network Layer Data link Layer Physical Layer Transport Layer

5) Find the number of addresses in the range if the first address is 152.79.52.30 and the last address is 152.79.56.78 (0.5)

52 53 1073 1072

6) What is the result of performing a bitwise AND operation between the IP address 29.135.14.56 and the subnet mask 255.255.140.0 (0.5)

29.135.12.0 29.135.14.0 29.135.12.56 29.135.14.56

7) Find the class of the following IPv4 address: 11000010 10011011 10101001 11111111 (0.5)

Class A Class B Class C Class D

8) In the classful addressing if the network is divided into 8 subnetworks and if one of the addresses in one of the subnets is 141.14.120.77, then the subnet address is: (0.5)

141.14.255.255 141.14.96.00 141.14.0.0 141.14.120.0

9) Which of the following does not have a Net ID and Host ID? (0.5)

Class A Class B Class C Class D

10) If you put all binary 1's in host identifier in an IP address, you get the _____ (0.5)

broadcast address | first valid IP address | last valid IP address | network address

## DESCRIPTIVE

**Answer all the questions.**

11) In classful addressing, an organization is granted the block 16.0.0.0/8. The administrator wants to create 8 fixed-length subnets. Analyze the block and determine (4)

a. The subnet mask.
b. The number of addresses in each subnet.
c. The first and the last address in the first subnet.
d. The first and the last address in the last subnet

.

12) A sender needs to transmit a total of 800,000 bits of data using the Stop-and-Wait ARQ protocol, where each frame carries 1000 bits. The transmission can take place over two possible physical links. Link A has a one-way distance of 400 km, a propagation speed of $2 \times 10^8$ m/s, and a transmission time per frame of 1 millisecond. Link B has a one-way distance of 100 km, the same propagation speed of $2 \times 10^8$ m/s, but a transmission time per frame of 2 milliseconds. For both links, assume that acknowledgment transmission time and processing delays are negligible, and that no frames are lost or corrupted. (3)

a. Calculate the total time required to send all the data over each link.
b. Based on your results, analyze which link is faster and explain the tradeoff observed between propagation delay and transmission time.

13) (3)

A communication engineer is tasked with designing a digital modulation scheme for a telemetry system. The available frequency band is from 200 kHz to 400 kHz. To efficiently transmit data, the system must encode 3 bits per symbol using Multiple Frequency Shift Keying (MFSK).

As the engineer, determine:

a. How many distinct frequency levels should be used.

b. The symbol (baud) rate of the system.

c. The approximate bandwidth requirement for this design.

.

14) Eight voice channels, each occupying a bandwidth of 3.2 kHz, are to be transmitted over a link of 25.6 kHz available in the range 140.0 to 165.6 kHz. The channels are first combined into two TDM frames, each carrying four voice channels, and then the two frames are multiplexed into the link using FDM without guard bands. Show, using the frequency domain, how the sub-bands are allocated and discuss how TDM and FDM together allow all eight channels to be transmitted simultaneously. (3)

15) Apply the CRC method to the following data:

Message D = 1101101010 and Generator Polynomial P = 101011

a. Determine the transmitted message.
b. Verify correctness of the received message assuming no transmission error. (3)

.

16) Compare circuit switching and packet switching in terms of setup and resource usage. Further, calculate the total transmission delay when a 2000-bit packet traverses 3 store-and-forward switches connected with 1 Mbps links. (3)

17) A 1000-byte packet is sent from host A to host B through two routers (R1 and R2) over three store-and-forward links, each operating at 1 Gbps and having a propagation delay of 2 ms. Ignoring processing delays, calculate the total time taken for the last bit of the packet to reach host B. (2)

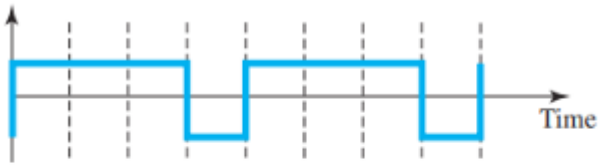
.

18) Find the class of the following IP address. Also find the first address and last address in each block .

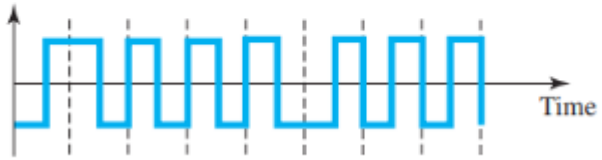
a. 15.25.14.100

b. 130.45.67.89 (2)

19) Find the 8-bit data stream for given figure. (2)




a. NRZ-I




b. differential Manchester